Lecture 1

Descriptive Regression with a Binary Covariate

Outline

- Univariate Descriptive Statistics
- Descriptive Regression with a Binary Covariate

What are descriptive statistics?

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- Having the whole population is becoming more common in the era of Big Data.
- Note: if you are interested in the causal effects of a treatment (a drug, a policy, etc.), you can never really have the whole population (we will discuss this in a few weeks).

What makes a good summary?

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From the google definition:

1. brief
2. includes main points

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   - minimizes the information lost by reducing the data
     - I'm using "information" in a loose sense here, will formalize a bit later
     - reducing the data means taking the data, which are often represented by $n * v$ numbers (with n the number of observations and v the number of variables) and reducing to a descriptive statistic that is represented by a few numbers.

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Enough forest. Trees (examples) are coming!

"Hold on, where's the forest again?"

It is easy in a statistics class to focus on the specifics. I want you to remember why we are doing what we are doing.

Outline

Univariate Descriptive Statistics

Univariate Example: SCOTUS data

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Univariate Example: SCOTUS data

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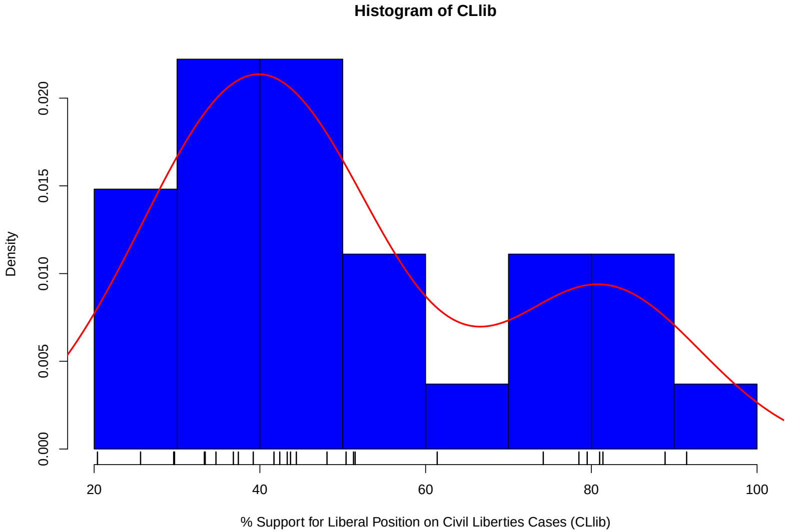
- Data can be interpreted as a census for the 1953 - 2005 era.
- One Variable: Percentage of votes in liberal direction for each justice in civil liberties cases

Justice	CLlib
Black	74.2
Reed	33.3
Thomas	25.6

Describing Univariate Data



Description/Summarization

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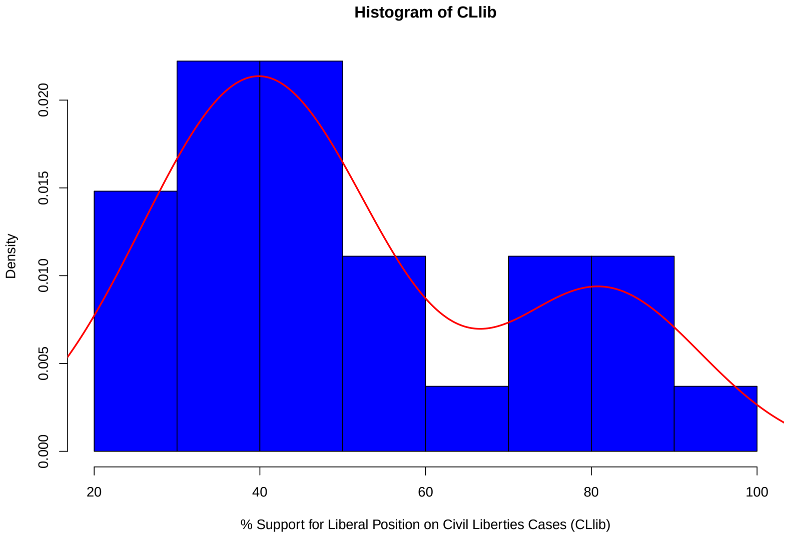
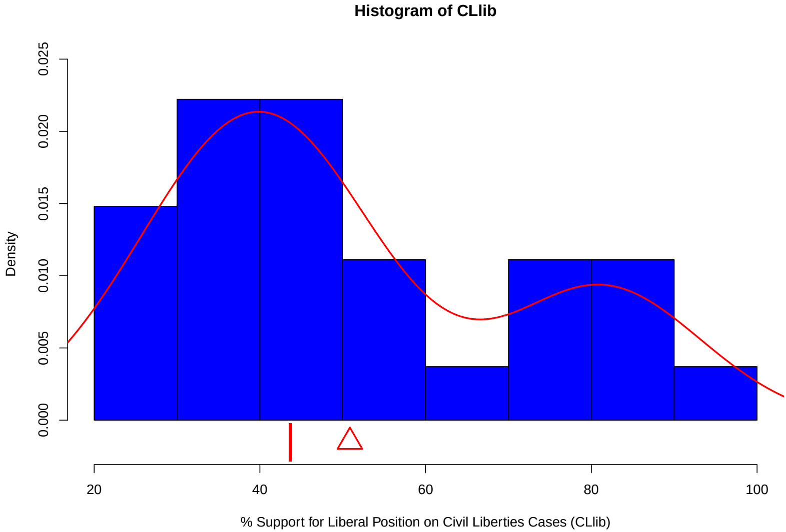
- We typically focus on measures of center and spread.
- We will discuss the rationale for this over the coming weeks.

Univariate Activity



Find the rough location of the mean and median.

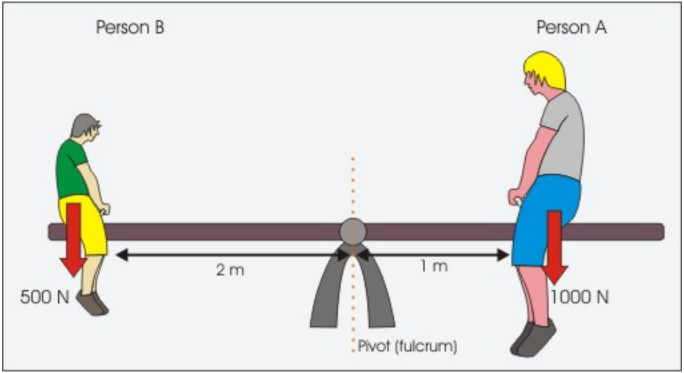
Univariate Activity Solution



Mean/Median Discussion

In pairs, discuss what the mean and median represent in the context of the CLib variable for these justices. What does it mean for the mean to be larger than the median in this context?

Mean as a balance point



Choosing a Descriptive Statistic

How might we choose between these descriptive statistics?

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A Least Squares Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y, by minimizing the sum of squared residuals (i.e., least squares).

$$SSR(\tilde{m}) = \sum_{i=1}^{N}(y_i - \tilde{m})^2$$

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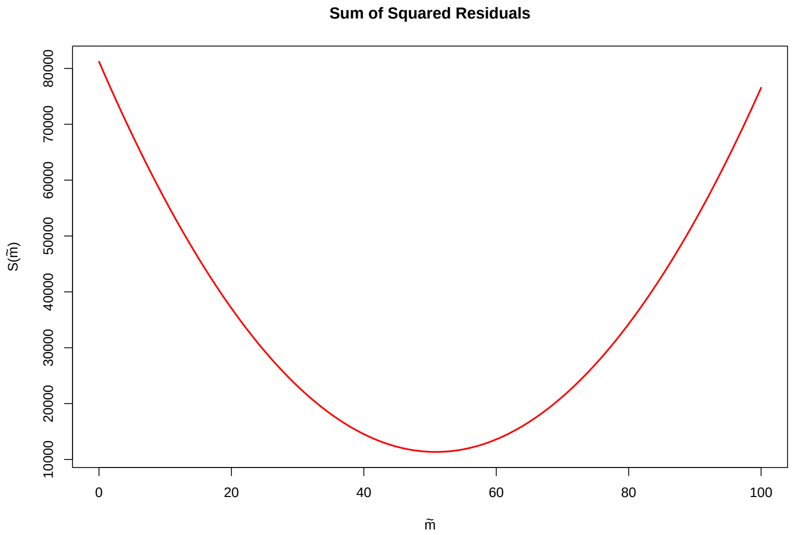
Activity: Using calculus, derive the least squares estimator

1. Calculate the derivative of SSR with respect to $\tilde{m}$
2. Set the derivative equal to 0
3. Solve for m

The Objective Function for SSR CLlib



1.

$$
\begin{aligned}
SSR(\tilde{m}) &= \sum_{i=1}^{N}(y_i - \tilde{m})^2 \\
&= \sum_{i=1}^{N}(y_i^2 - 2y_i\tilde{m} + \tilde{m}^2)
\end{aligned}
$$

$$
\frac{\partial SSR(\tilde{m})}{\partial \tilde{m}} = \sum_{i=1}^{N}(-2y_i + 2\tilde{m})
$$

The Slope of the Tangent Line for SSR CLlib

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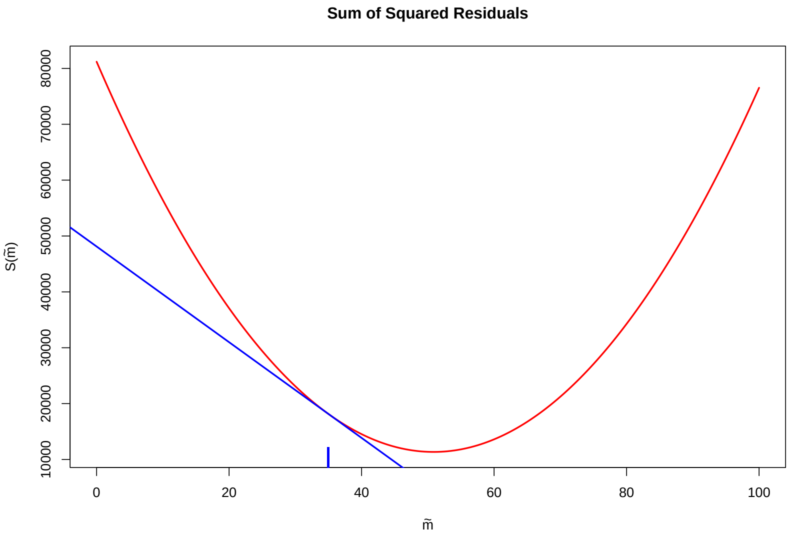

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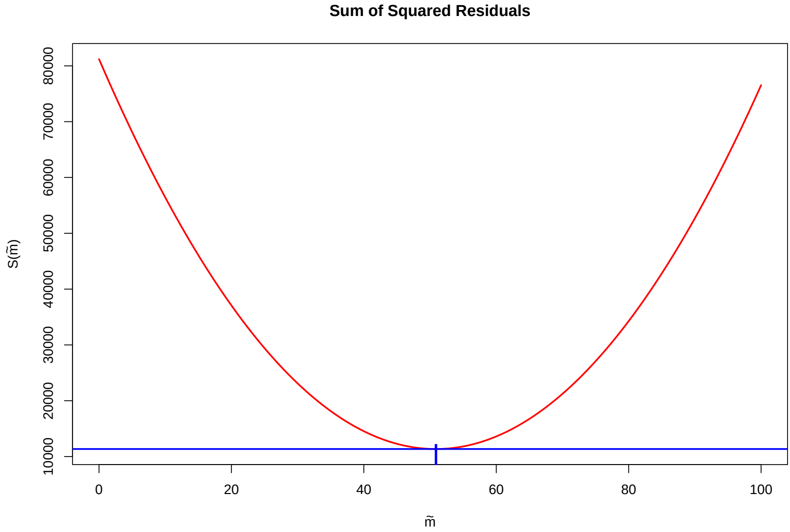

1.

$$SSR(\tilde{m}) = \sum_{i=1}^{N} (y_i - \tilde{m})^2$$
$$= \sum_{i=1}^{N} (y_i^2 - 2y_i\tilde{m} + \tilde{m}^2)$$
$$\frac{\partial S(m)}{\partial \tilde{m}} = \sum_{i=1}^{N} (-2y_i + 2\tilde{m})$$

2.

$$0 = \sum_{i=1}^{N} (-2y_i + 2m)$$

3.

$$m = \frac{1}{N} \sum_{i=1}^{N} y_i$$

Therefore, the mean is a least squares summary.

An Least Absolute Value Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y, by minimizing the sum of absolute deviations.

$$SAV(\tilde{m}) = \sum_{i=1}^{N} |y_i - \tilde{m}|$$

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- Why is this harder to solve?
- Any guesses?

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- The properties of means are relatively easy to describe.
- Randomized experiments only identify mean causal effects (as do regression and matching when the appropriate assumptions hold).

Variance and Standard Deviation

- Mean is the least squares predictor in terms of SSR.
- How small is “least” (i.e., how much spread around the mean)?
- Variance and Standard Deviation are useful transformations of SSR for the mean.

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Variance

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\begin{aligned}
S^2 &= \frac{\sum_{i=1}^{N}(y_i - \bar{y})^2}{N-1} \\
&= \frac{SSR}{N-1}
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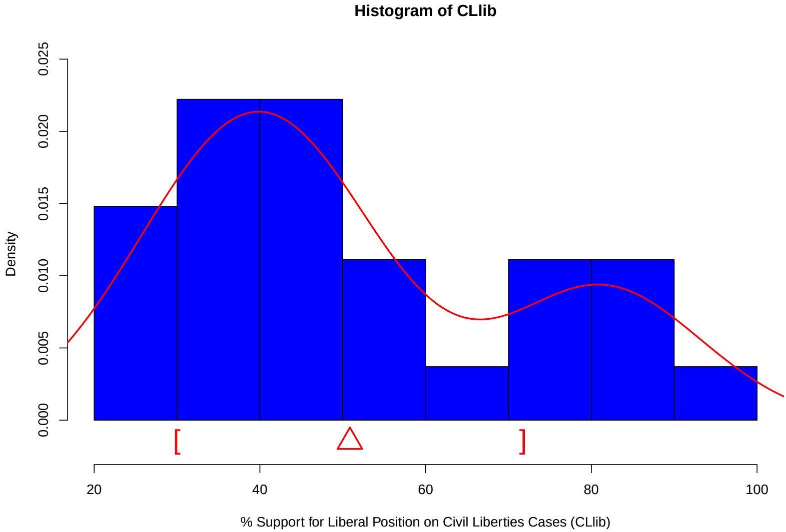
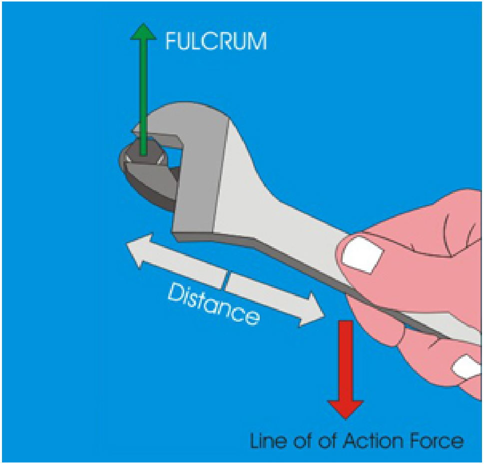
Standard Deviation

$$S = \sqrt{S^2}$$

Mean and Standard Deviation



Variance as turning force



Why Variance and Standard Deviation?

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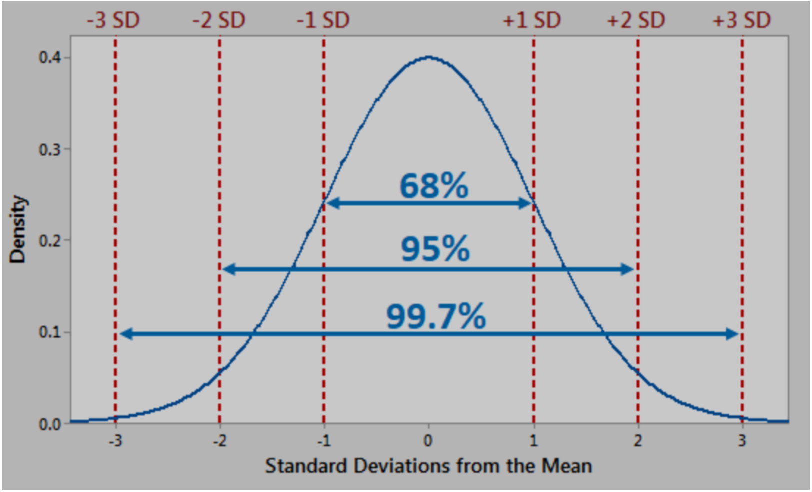
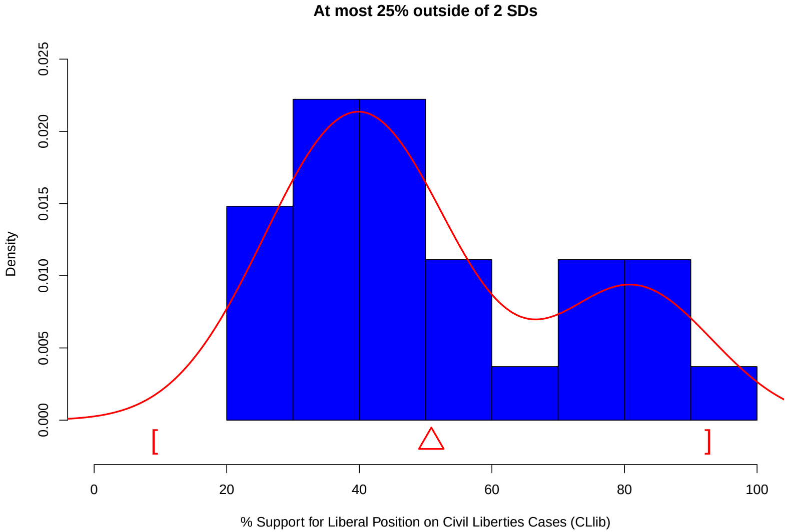
- Means and standard deviations are sufficient to describe the distribution of normally distributed data.
- For non-normal data, means and standard deviations provide information via Chebyshev's inequality and other concentration inequalities.

Normally-Shaped Histograms



Chebyshev for Non-Normal Histograms



Special Case: Binary Variables

Justice	Underrepresented Group
Black	0
Marshall	1
Burger	0
O'Connor	1
Scalia	0
Thomas	1
Proportion	3/27

Mean and Variance for UR Group Variable

$$\begin{aligned}
\bar{y} &= \frac{1}{N}\sum_{i=1}^{N} y_i \\
&= \frac{0+\cdots+1+0+\cdots+1+0+\cdots+1}{27} \\
&= \frac{3}{27}
\end{aligned}$$

$$\begin{aligned}
S^2 &= \frac{\sum_{i=1}^{N}(y_i-\bar{y})^2}{N-1} \\
&= \frac{(0-.11)^2+\cdots+(1-.11)^2}{27-1} \\
&= \left(\frac{27}{27-1}\right)\left[\bar{y}\cdot(1-\bar{y})\right]
\end{aligned}$$

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Descriptive Regression with a Binary Covariate

Conditional Means

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- All additional variables adjust how we consider the outcome, often by conditioning.

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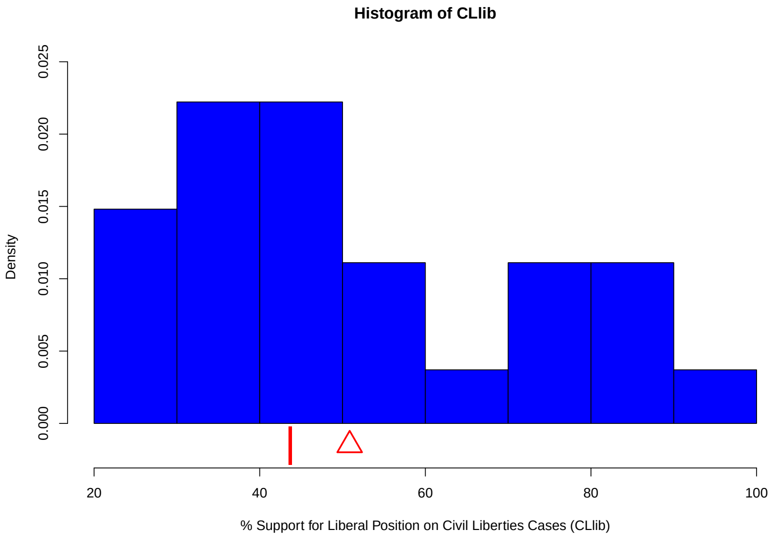
- All additional variables adjust how we consider the outcome, often by conditioning.
- Typically, we consider the mean of y conditional on values of other variables (called covariates).
- Consider the binary covariate Party (Rep 0, Dem 1).

Justice	CLlib	Party
Black	74.2	1
Reed	33.3	1
Thomas	25.6	0

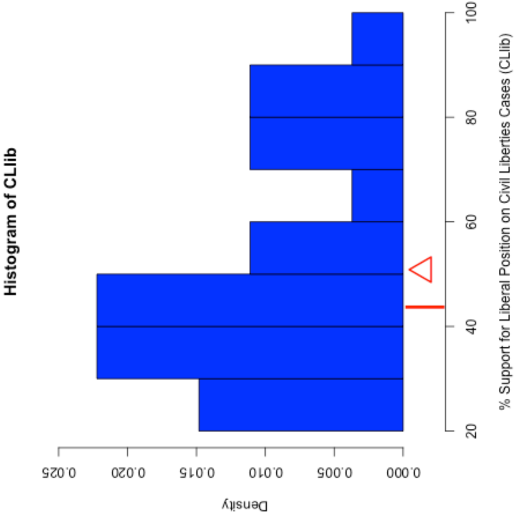
From Univariate to Regression



From Univariate to Regression

pdf

2



Binary Covariate Regression

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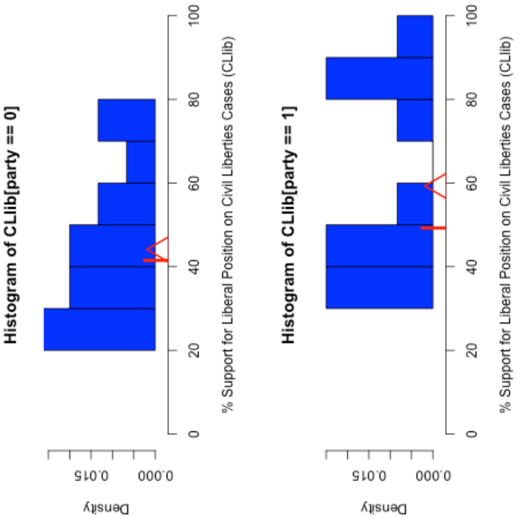
2

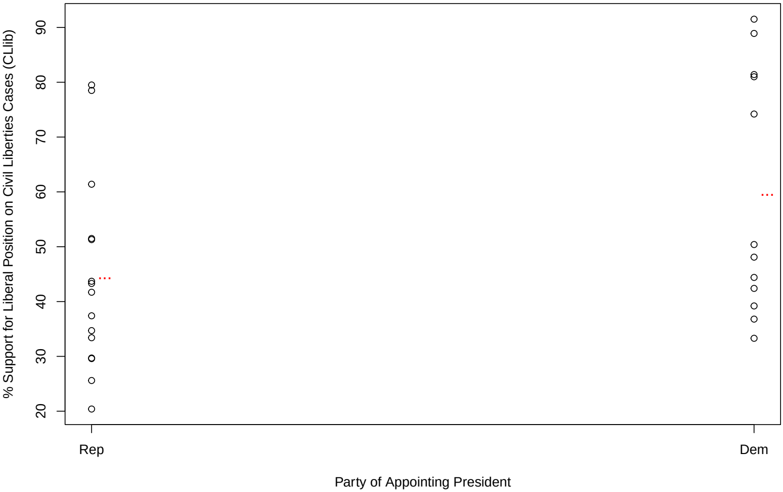


Conditional Mean/Median Discussion

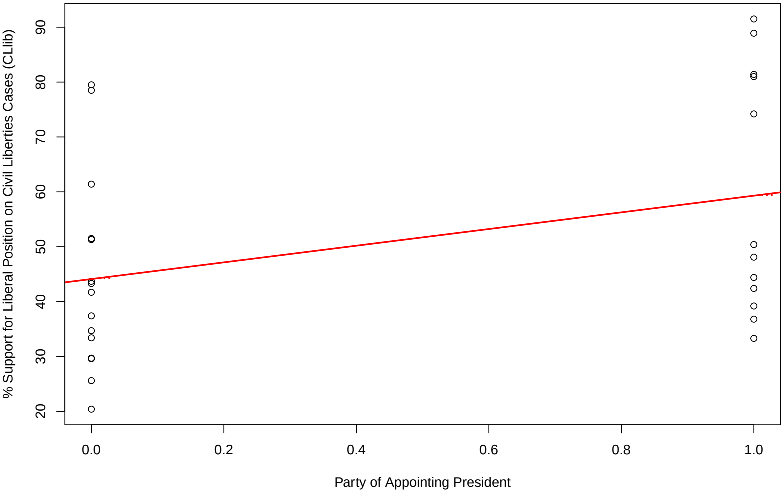
In pairs, discuss ...

1. What do the mean and median represent in the context of the CLib variable for the justices of each party?
2. What does it mean for the mean to be larger for the Democrats than the Republicans?
3. What does it mean for the difference in means to be larger than the difference in medians?

Scatterplot Representation



OLS Regression

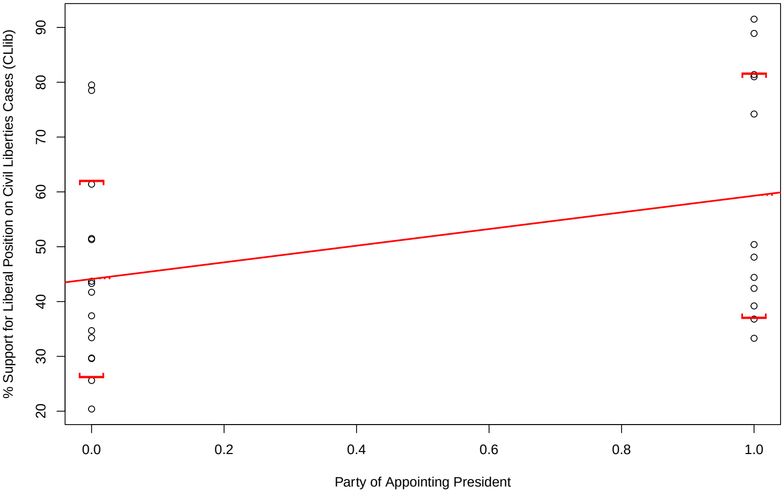


Intercept and Slope Discussion

In pairs, discuss ...

1. The substantive interpretation of the intercept and slope of the line in terms of justices for each party.
2. What does it mean for the mean to be larger for the Democrats than the Republicans?
3. What does it mean for the difference in means to be larger than the difference in medians?

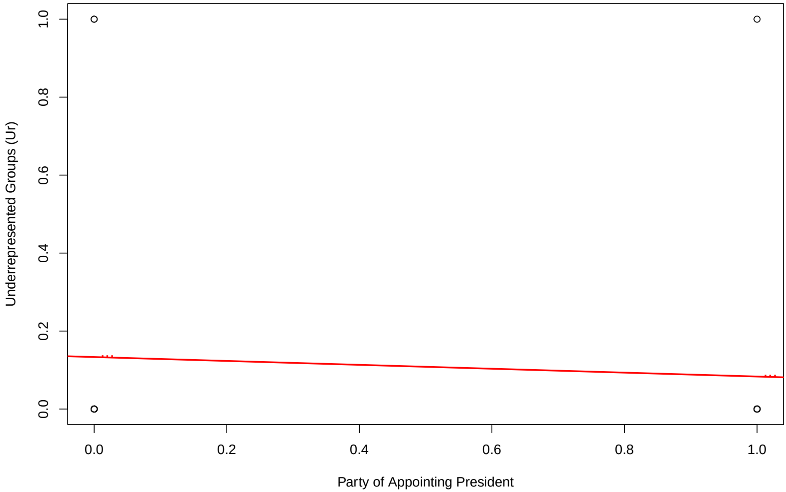
Conditional Standard Deviations



Special Case: Binary Outcomes

Justice	Ur	Party
Black	0	1
Reed	0	1
Thomas	1	0

Binary Outcome OLS Regression



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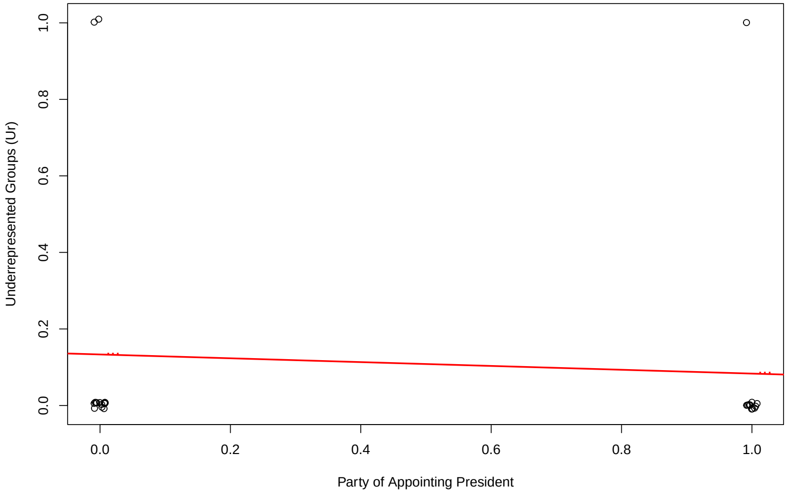
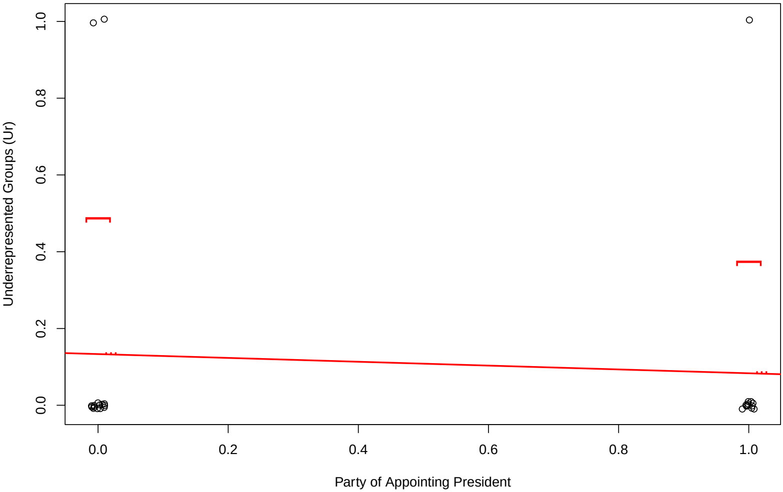


Conditional Standard Deviations



Binary Outcome OLS Discussion

In pairs, discuss ...

1. The substantive interpretation of the intercept and slope of the line in terms of justices for each party.
2. Why is the SD smaller for Democrats than Republicans (hint: look at the variance formula for binary variables)

Summary